

Lab assignment 3 - Edge detection, Image resampling, and Template matching

Complete

22.9.2025

Attempt 1



Review Feedback

Attempt 1 Score:

15.9.2025

Complete



Add Comment

Anonymous Grading: no

Unlimited Attempts Allowed

8.9.2025 to 17.11.2025

Details

(<https://www2.eecs.berkeley.edu/Research/Projects/CS/vision/bsds/BSDS300/html/dataset/images/gray/test-001-025.html>)

Lab assignment 3 (individual)

In this individual lab assignment, every student must complete the following tasks.

The code created for the exercises has to be delivered to the repository from Lab assignment 1.

SUBMISSION: The submission shall be a link to your git repository along with the outputs from part II.

In this exercise, we will use the image "[lambo.png](https://uia.instructure.com/courses/18389/files/3005660?wrap=1) (<https://uia.instructure.com/courses/18389/files/3005660?wrap=1>)".  (https://uia.instructure.com/courses/18389/files/3005660/download?download_frd=1)". Use "cv2.imread()" to load the image!

I. Assignment directory

1. Create a new project with PyCharm:
 - A. Create a new directory named "assignment_3" inside your "IKT213_lastname" repository.

II. Create functions for edge detection, resampling, and template matching

1. **Sobel edge detection:** Your task is to detect the edges of the model Lamborghini using Sobel filters.

To learn more about Sobel edge detection, go to "<https://learnopencv.com/edge-detection-using-opencv/>" (<https://learnopencv.com/edge-detection-using-opencv/>)".

- A. Function name: "sobel_edge_detection"
- B. Function argument: **image**
- C. Use **GaussianBlur()** with **ksize=(3,3)**, and **SigmaX=0**

- D. Apply **Sobel edge detection** to the blurred image, use **dx=1, dy=1, ksize=1**
- E. Save the image

2. **Canny edge detection: Your task is to detect the edges of the model Lamborghini using Canny filters**

to learn more about Canny edge detection, go to "https://opencv24-python-tutorials.readthedocs.io/en/latest/py_tutorials/py_imgproc/py_canny/py_canny.html" (https://opencv24-python-tutorials.readthedocs.io/en/latest/py_tutorials/py_imgproc/py_canny/py_canny.html).

- A. Function name: **"canny_edge_detection"**
- B. Function arguments: **image, threshold_1, threshold_2**
- C. Use **GaussianBlur()** with **ksize=(3,3)**, and **SigmaX=0**
- D. Apply **Canny edge detection** to the blurred image (use **threshold_1=50** and **threshold_2=50**)
- E. Save the image

3. (https://docs.opencv.org/4.5.1/d4/d1f/tutorial_pyramids.html) **Template match: Your task is, for a given image and template, you must locate any segment of the image that matches what is given from the template**

To learn more about template matching, go to "https://opencv24-python-tutorials.readthedocs.io/en/latest/py_tutorials/py_imgproc/py_template_matching/py_template_matching.html" (https://opencv24-python-tutorials.readthedocs.io/en/latest/py_tutorials/py_imgproc/py_template_matching/py_template_matching.html).

Use image "[shapes.png](https://uia.instructure.com/courses/18389/files/3028137?wrap=1)" (<https://uia.instructure.com/courses/18389/files/3028137?wrap=1>) and the template match image "[shapes_template.jpg](https://uia.instructure.com/courses/18389/files/3028138?wrap=1)" (<https://uia.instructure.com/courses/18389/files/3028138?wrap=1>)

- A. Function name: **"template_match"**
- B. Function argument: **image, template**
- C. Use **matchTemplate()**, set the **threshold=0.9**, and mark the matched areas with a **red** rectangle like "Template Matching with Multiple Objects" from the link above.
(Hot tip 🔥 : both the image and the template must be in gray **only** to perform the matching)
- D. Save the image

4. **Resizing: Your task is to "zoom in/out" a given image using image pyramids**

to learn more about resizing, go to "https://docs.opencv.org/4.5.1/d4/d1f/tutorial_pyramids.html" (https://docs.opencv.org/4.5.1/d4/d1f/tutorial_pyramids.html).

- A. Function name: **"resize"**
- B. Function arguments: **image, scale_factor: int, up_or_down: str**
- C. The variable **scale_factor=2** is an int and the variable **up_or_down** is a string: "up" or

"down"

D. Resize the image using the string to resize up or down with the given scale_factor

E. Save the image

Do you need help in the lab or in Discord? Find links here: [Lab Information \(https://uia.instructure.com/courses/18389/pages/lab-information\)](https://uia.instructure.com/courses/18389/pages/lab-information)
